Math 510 **Fall 2026**
Homework 4 **Drew Armstrong**

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1. Trivial Kernel. Prove that a linear function f is injective if and only if $ker(f) = \{\mathbf{0}\}$. Prove that the columns of a matrix A are independent if and only if $\mathcal{N}(A) = \{\mathbf{0}\}$.

2. Rank-Nullity Theorem. Let U and V be finite-dimensional vector spaces, and let $f : U \to V$ be a linear function. Prove that

$$\dim im(f) + \dim ker(f) = \dim U.$$

[Hint: Let $\mathbf{u}_1, \ldots, \mathbf{u}_k$ be a basis for $ker(f)$, and extend this to a basis $\mathbf{u}_1, \ldots, \mathbf{u}_{k+1}, \ldots, \mathbf{u}_n$ for U. Prove that $f(\mathbf{u}_{k+1}), \ldots, f(\mathbf{u}_n)$ is a basis for $im(f)$.]

3. Intersection of Linear Hyperplanes. A *linear hyperplane* is a vector subspace $H \subseteq V$ of dimension $\dim V - 1$. Suppose that V is also an inner product space.

(a) Prove that every linear hyperplane $H \subseteq V$ has the form

$$H = \mathbf{u}^\perp = \{\mathbf{x} \in V : \langle \mathbf{u}, \mathbf{x} \rangle = 0\},$$

for some unit vector $\mathbf{u} \in V$. [Hint: Consider the orthogonal complement $H^\perp \subseteq V$.]

(b) Given linear hyperplanes $H_i = \mathbf{u}_i^\perp$, prove that

$$H_1 \cap H_2 \cap \cdots \cap H_m = \mathrm{Span}\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}^\perp.$$

(c) Express part (b) in terms of the nullspace and the row space of some matrix.

4. Affine Hyperplanes. An *affine hyperplane* is an affine subspace $H \subseteq V$ of dimension $\dim V - 1$. If V is a real inner product space then one can show using the previous problem that any affine hyperplane H can be expressed as

$$H = H_{\mathbf{u},k} = \{\mathbf{x} \in V : \langle \mathbf{x}, \mathbf{u} \rangle = k\}$$

for some unit vector $\mathbf{u} \in V$ and scalar $k \in \mathbb{R}$. Let $P_{\mathbf{u},k} : V \to V$ denote the orthogonal projection onto $H_{\mathbf{u},k}$ and let $F_{\mathbf{u},k} : V \to V$ denote the orthogonal reflection across $H_{\mathbf{u},k}$.

(a) Fix any point $\mathbf{x}' \in H_{\mathbf{u},k}$, so that $\langle \mathbf{x}', \mathbf{u} \rangle = k$. Show that we can write

$$H_{\mathbf{u},k} = \mathbf{u}^\perp + \mathbf{x}' = \{\mathbf{x} + \mathbf{x}' : \mathbf{x} \in \mathbf{u}^\perp\}.$$

(b) By translating to the origin, one can show that

$$P_{\mathbf{u},k}(\mathbf{x}) = P_{\mathbf{u},0}(\mathbf{x} - \mathbf{x}') + \mathbf{x}',$$
$$F_{\mathbf{u},k}(\mathbf{x}) = F_{\mathbf{u},0}(\mathbf{x} - \mathbf{x}') + \mathbf{x}'.$$

Use this to find explicit formulas for $P_{\mathbf{u},k}(\mathbf{x})$ and $F_{\mathbf{u},k}(\mathbf{x})$. [Hint: Recall that $P_{\mathbf{u},0}(\mathbf{x}) = \mathbf{x} - \langle \mathbf{u}, \mathbf{x} \rangle \mathbf{u}$ and $F_{\mathbf{u},0}(\mathbf{x}) = \mathbf{x} - 2\langle \mathbf{u}, \mathbf{x} \rangle \mathbf{u}$.]

(c) Compute the projection of the origin onto $H_{\mathbf{u},k}$.

(d) Describe the composition $F_{\mathbf{u},k} \circ F_{\mathbf{u},\ell}$ of two parallel reflections. See the challenge problem for the composition of two non-parallel reflections.

5. Left Inverses. Let $A \in \mathbb{C}^{m \times n}$ have independent columns, so that $m - n \geq 0$. In this problem you will find a bijection between $\mathbb{C}^{n \times (m-n)}$ and the set of left inverses of A.

(a) Prove that $\mathcal{N}(A^*A) = \mathcal{N}(A)$. [Hint: Show that $\mathbf{x} \in \mathcal{N}(A^*A)$ implies $\|A\mathbf{x}\| = 0$.]

(b) Use (a) to prove that A^*A is invertible. Show that

$$L := (A^*A)^{-1}A^*$$

is a left inverse of A.

(c) Let $N \in \mathbb{C}^{m\times(m-n)}$ have columns forming an orthonormal basis of $\mathcal{N}(A^*)$, and consider any matrix $X \in \mathbb{C}^{n\times(m-n)}$. Define the matrix

$$L_X := L + XN^*.$$

Show that L_X is a left inverse of A. [Hint: Show that $N^*A = O$.]

(d) Prove that every left inverse of A has the form L_X for some unique $X \in \mathbb{C}^{n\times(m-n)}$. [Hint: Let L' be any left inverse of A and consider $M = L' - L$. Show that $A^*M^* = O$, so every column of M^* belongs to $\mathcal{N}(A^*)$. It follows that $M^* = NY$ for some matrix Y. For uniqueness, use the fact that $N^*N = I$.]

(e) What happens in the case $m = n$? [Idea: Maybe just define $\mathbb{C}^{n\times 0} := \emptyset$.]

Challenge Problem (Euler-Rodrigues Formula). For any unit vector $\mathbf{u} \in \mathbb{R}^n$, recall that $F_\mathbf{u} = I - 2\mathbf{u}\mathbf{u}^T$ is the matrix that reflects across the hyperplane $\mathbf{u}^\perp$. Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ be unit vectors with angle α between them (assume $0 < \alpha < \pi$) and let $U = \text{Span}\{\mathbf{u}, \mathbf{v}\} \subseteq \mathbb{R}^n$ be the plane that they span. We will show that the product matrix $R := F_\mathbf{u}F_\mathbf{v}$ fixes $U^\perp$ and rotates U by angle 2α. To simplify notation we write $c := \cos\alpha$ and $s := \sin\alpha$. The key idea of the proof is to consider the matrix

$$J := \frac{1}{s}(\mathbf{u}\mathbf{v}^T - \mathbf{v}\mathbf{u}^T).$$

We will see that J vanishes on $U^\perp$ and performs a quarter turn on U.

(a) Show that $\mathbf{x} \in U^\perp$ implies $J\mathbf{x} = \mathbf{0}$.

(b) For $\mathbf{x} \in U$, show that $J^2\mathbf{x} = -\mathbf{x}$. [Hint: Show that $J^2\mathbf{u} = -\mathbf{u}$ and $J^2\mathbf{v} = -\mathbf{v}$.]

(c) For $\mathbf{x} \in U$, show that $J\mathbf{x} \in U$, $\|J\mathbf{x}\| = \|\mathbf{x}\|$ and $\langle J\mathbf{x}, \mathbf{x}\rangle = 0$. In other words, J rotates the plane U by 90°. [Hint: Use (b) and the fact that $J^T = -J$.]

(d) Show that

$$\boxed{R = I + \sin(2\alpha)J + (1 - \cos(2\alpha))J^2}.$$

This is called the *Euler-Rodrigues formula* for rotation matrices. [Hint: Use the trig identities $\sin(2\alpha) = 2cs$ and $1 - \cos(2\alpha) = 2s^2$.]

(e) For $\mathbf{x} \in U$, show that

$$R\mathbf{x} = \cos(2\alpha)\mathbf{x} + \sin(2\alpha)J\mathbf{x}.$$

Explain why this shows that R rotates the plane U by angle 2α.

(f) For any square matrix, one can define the *matrix exponential*

$$\exp(A) := I + A + \frac{1}{2}A^2 + \frac{1}{6}A^3 + \cdots = \sum_{k\geq 0}\frac{1}{k!}A^k,$$

which converges with respect to any matrix norm. Use the Euler-Rodrigues formula to show that

$$\boxed{R = \exp(2\alpha J)}.$$

The matrix J is called an *infinitesimal rotation*. [Hint: Parts (a) and (b) imply that $J^3 = -J$, so that $J^{2r+1} = (-1)^r J$ and $J^{2r+2} = (-1)^r J^2$ for all integers $r \geq 0$. Use the Taylor series for sin and cos.]